

STAT/MA 41600  
In-Class Problem Set #19: October 2, 2026  
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**Problem Set 19 Answers**

**1.** Suppose  $X$  is a Hypergeometric random variable with parameters  $n = 5$ ,  $M = 20$ , and  $N = 30$ .

**1a.** We have  $P(X = 3) = \binom{20}{3}\binom{10}{2}/\binom{30}{5} = 950/2639 = 0.35998$ .

**1b.** The random  $X$  can be approximated by a Binomial random variable  $Y$  with  $n = 5$  and  $p = M/N = 20/30 = 2/3$ . We have  $P(Y = 3) = \binom{5}{3}(2/3)^3(1/3)^2 = 0.3292$ .

**2.** Suppose  $X$  is a Binomial random variable with parameters  $n = 7$  and  $p = 1/2$ . Suppose  $Y$  is a Hypergeometric random variable with parameters  $n = 35$ ,  $M = 20$ , and  $N = 200$ .

We have  $\text{Var}(X) = np(1 - p) = (7)(1/2)(1/2) = 7/4 = 1.75$  and also  $\text{Var}(Y) = (n)(M/N)(1 - M/N)(N - n)/(N - 1) = (35)(20/200)(1 - 20/200)(200 - 35)/(200 - 1) = 2079/796 = 2.6118$ .

**3a.** The probability that she selects an equal number of green jelly beans and red jelly beans (i.e., 4 jelly beans of each color) is  $\binom{300}{4}\binom{200}{4}/\binom{500}{8} = 0.2336$

**3b.** The expected number of green jelly beans Katie selects is  $(8)(300/500) = 24/5 = 4.8$ .

**3c.** The variance of the number of green jelly beans Katie selects is

$$(8)(300/500)(1 - 300/500)(500 - 8)/(500 - 1) = 23616/12475 = 1.8931.$$

**4.** The probability mass function of the number  $X$  of green jelly beans that Katie selects is  $p_X(x) = \binom{300}{x}\binom{200}{8-x}/\binom{500}{8}$  for  $0 \leq x \leq 8$ . This yields:

$$p_X(0) = 0.0006 \quad p_X(1) = 0.0075 \quad p_X(2) = 0.0404 \quad p_X(3) = 0.1234$$

$$p_X(4) = 0.2336 \quad p_X(5) = 0.2809 \quad p_X(6) = 0.2092$$

and we do not really need  $p_X(7)$  or  $p_X(8)$ , for this question!

The probability mass function of the number  $Y$  of green balls that Matthias selects is  $p_Y(y) = \binom{25}{y}\binom{15}{7-y}/\binom{40}{7}$  for  $0 \leq y \leq 7$ . This yields:

$$p_Y(1) = 0.0067 \quad p_Y(2) = 0.0483 \quad p_Y(3) = 0.1684 \quad p_Y(4) = 0.3087$$

$$p_Y(5) = 0.2992 \quad p_Y(6) = 0.1425 \quad p_Y(7) = 0.0258$$

and we do not really need  $p_Y(0)$  for this question.

Finally, the desired probability is  $\sum_{x=0}^6 \sum_{y=x+1}^7 p_X(x)p_Y(y) = 0.3038$ .

**5.** The probability mass function of  $X$  is  $p_X(x) = \binom{10}{x}\binom{15}{3-x}/\binom{25}{3}$  for  $0 \leq x \leq 3$ .

$$p_X(0) = 91/460 = 0.1978 \quad p_X(1) = 21/46 = 0.4565$$

$$p_X(2) = 27/92 = 0.2935 \quad p_X(3) = 6/115 = 0.0522$$